

Periodic Scalar Play as a Sweeping Process: Exact Clamp, Entrainment, and Dissipation

Route-A source-local certificate HCS-C332

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Abstract

For a scalar Moreau play element driven by one periodic rise and fall, we prove the *segment projection and exact Poincaré clamp* for every $W^{1,1}$ input, including plateaus and corners. The *three chambers and one-period entrainment* follow from one idempotent map, together with order, nonexpansion, and rate independence. We close *variation, dissipation, and Route-A obstruction*, including $r = 0$, constant input, equality, finite hostile evidence, and source limits.

1 Segment solution and return map

Fix $r \geq 0$. Let $u \in W^{1,1}([0, T])$ satisfy $u(0) = u(T) = m$. For some τ , assume that u is nondecreasing from m to M on $[0, \tau]$ and nonincreasing back to m on $[\tau, T]$. Plateaus are allowed. Put $D = M - m$ and

$$C(t) = [u(t) - r, u(t) + r].$$

We use $N_C(y) = \{v : v(z - y) \leq 0 \text{ for all } z \in C\}$ and solve

$$y(t) \in C(t), \quad -\dot{y}(t) \in N_{C(t)}(y(t)) \quad \text{a.e.} \quad (1)$$

Theorem 1 (Projection and Poincaré clamp). *For every $z \in [m - r, m + r]$, problem (1) with $y(0) = z$ has a unique absolutely continuous solution. On a monotone segment beginning at t_0 ,*

$$y(t) = \text{Proj}_{[u(t)-r, u(t)+r]} y(t_0). \quad (2)$$

The return map from one minimum to the next is

$$P(z) = \min\{m + r, \max\{M - r, z\}\}. \quad (3)$$

Proof. On a nondecreasing segment define $y(t) = \max\{y(t_0), u(t) - r\}$. It is feasible. Before the lower endpoint reaches the state, $\dot{y} = 0$; when the endpoint pushes it, $y = u - r$, $\dot{y} = \dot{u} \geq 0$, and $-\dot{y}$ lies in the lower-endpoint normal cone. Feasibility and the normal-cone sign force these alternatives for any solution, proving uniqueness. On a nonincreasing segment the corresponding formula is $y(t) = \min\{y(t_0), u(t) + r\}$. Both formulas are (2) and remain valid almost everywhere across plateaus and corners.

The rise ends at $z_\uparrow = \max\{z, M - r\}$. The fall ends at $\min\{z_\uparrow, m + r\}$, proving (3). $\square$

2 Three chambers and finite-time entrainment

Theorem 2 (Complete periodic atlas). *The map P is increasing, one-Lipschitz, and idempotent. Every admissible initial state therefore reaches a periodic response in at most one period. Its fixed responses are as follows.*

1. If $D < 2r$, the fixed set is $[M - r, m + r]$, and each fixed response is constant.
2. If $D = 2r$, this interval collapses to the single constant response $m + r = M - r$.
3. If $D > 2r$, the unique fixed state is $m + r$ and it generates the unique nonconstant periodic loop.

For a common input, the solution maps preserve initial order and satisfy $|y_1(t) - y_2(t)| \leq |y_1(0) - y_2(0)|$.

Proof. When $D < 2r$, equation (3) is scalar projection onto $[M - r, m + r]$; at equality the interval is a singleton. When $D > 2r$, $M - r > m + r$, so (3) is the constant $m + r$. This proves all fixed sets and directly gives $P^2 = P$. A fixed point in the first two chambers is never pushed through a positive distance. In the third chamber, it moves from $m + r$ to $M - r$ and back. Projection on an interval is increasing and one-Lipschitz; composing the two segment maps proves the last assertion. $\square$

If ϕ is an orientation-preserving absolutely continuous surjection and both u and $u \circ \phi$ lie in $W^{1,1}$, formula (2) depends only on the ordered input value. Uniqueness therefore gives

$$\mathcal{P}_r[u \circ \phi] = \mathcal{P}_r[u] \circ \phi.$$

This is the precise admissible reparameterization statement; no inadmissible composition is silently included.

3 Variation and dissipation

Set $s = u - y \in [-r, r]$. Almost everywhere, either $\dot{y} = 0$ or the moving endpoint pushes y and $\dot{s} = 0$. Hence

$$\text{Var}(u) = \text{Var}(y) + \text{Var}(s). \quad (4)$$

On an active rising lower endpoint, $s = r$ and $\dot{y} \geq 0$; on an active falling upper endpoint, $s = -r$ and $\dot{y} \leq 0$. Thus

$$\int_0^T s \, dy = r \, \text{Var}(y). \quad (5)$$

For any periodic response,

$$\text{Var}(y) = 2 \max\{D - 2r, 0\}, \quad \int_0^T s \, dy = 2r \max\{D - 2r, 0\}. \quad (6)$$

Indeed, the play is constant for $D \leq 2r$; above threshold it travels the distance $D - 2r$ once in each direction.

At $r = 0$, $C(t) = \{u(t)\}$, so $y = u$, $s = 0$, and every identity survives. At $D = 0$, C is fixed, P is the identity on $[m - r, m + r]$, and every feasible constant state is periodic. Equality $D = 2r$ has a unique constant response and zero dissipation. Plateau duration and $W^{1,1}$ corners do not change any formula.

4 Finite certificate and obstruction

The receipt contains 12 rational cases, all three chambers, every selected feasible initial state through two periods, coarse and plateau-stretched paths, and 893 audited scalar leaves. An independent checker performs 792 checks. SymPy closes 4,788 exact rational identities, two isolated replays are byte-identical, and 69 hostile parser, nested-schema, semantic, and repaired-hash attacks are rejected. These rows are regression evidence; Theorems 1–2 and equations (4)–(6) prove the full frozen $W^{1,1}$ result.

The nearest registered owners are C252, a discrete-guard relay oscillator; C238, forward Coulomb–Filippov capture; and C310, endpoint-controlled Dubins synthesis. None is a scalar convex moving-interval play law.

The conservative tuple is

$$(A0_FAIL, A1_PASS_ANALYTIC, \\ A2_FAIL, A3_FAIL, A4_FORMAL_HINT).$$

The exact periodic atlas has no rational-prime labels. The clamp is not an Euler product or target determinant, and the normal-cone dissipation law is not a natural unitary or self-adjoint quantization. Route A is rejected and Route B stays locked under `NO_BAD_EULER_OR_ROOT_NUMBER`. No target local data, root number, automorphy, divisor, functional equation, target-zero match, or Hilbert–Pólya operator is claimed.

AI use. A generative language model assisted drafting and code scaffolding. The proofs, independent recomputation, hostile tests, and deterministic artifacts define the audit record.

Source lineage

This source-local reconstruction makes no priority claim.

References

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- [2] M. A. Krasnosel’skii and A. V. Pokrovskii, *Systems with Hysteresis*, Springer, 1989. DOI: [10.1007/978-3-642-61302-9](https://doi.org/10.1007/978-3-642-61302-9).
- [3] M. Brokate and J. Sprekels, *Hysteresis and Phase Transitions*, Springer, 1996. DOI: [10.1007/978-1-4612-4048-8](https://doi.org/10.1007/978-1-4612-4048-8).